

# Citizen AI Dataset Infrastructure

*Pilot Strategy, Technical Stack, and Roadmap to World-Class Public-Good Datasets*

Standalone briefing document

Prepared for the Citizen AI / Abundance dataset team and external reviewers

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## Core thesis

Citizen AI is not trying to prove that non-experts can complete trivial labelling tasks. It is testing whether non-experts can become reliable contributors to real-world usable AI datasets through structured capture protocols, apprenticeship-style progression, spatial annotation workflows, review, quality measurement, and expert-derived training documentation.



## Document purpose

This document sets out a proposed delivery plan and technical architecture for Citizen AI dataset infrastructure. It is written as a standalone briefing for people who have not been part of the earlier conversations. It focuses specifically on the software and workflow needed to capture, label, review, quality-control, and release real-world usable datasets. It does not focus on the wider Kindness Platform except at the API boundary where identity, roles, task eligibility, and quality feedback need to pass between systems.

The immediate planning question is how to move from the current early tooling and volunteer capacity to a robust operational workflow by January or February 2027, using a summer adult pilot and an October school beta as learning stages. The proposed answer is to run human workflow pilots now, even with manual glue between tools, while a technical team builds the orchestration layer in parallel.

## Executive summary

- The foundational hypothesis is that non-experts can help build real-world usable datasets if they are trained through a structured progression path and supported by rigorous workflow infrastructure.
- The project should keep using KoboToolbox for field capture while it works at scale, but Kobo should be treated as a capture source, not the long-term dataset system of record.
- The project should build a Citizen AI Dataset Orchestrator using Django, PostgreSQL, Celery/Redis, and S3-compatible object storage. This orchestrator should own image custody, task queues, workflow state, event logging, review state, quality metrics, and release exports.
- The project should not make Label Studio, CVAT, Kobo, FiftyOne, or the Kindness Platform the source of truth for dataset workflow. These tools can be used for specific jobs, but the Orchestrator should own the canonical workflow

state.

- For simple and high-volume workflows, the project may eventually build a native labelling/review UI. For spatial annotation such as points, bounding boxes, polygons, segmentation, and video, CVAT should be integrated as a specialist annotation engine.
- The summer pilot should not wait for the full technical stack. It should use manual handoffs between Kobo, CVAT or Label Studio, spreadsheets, exports, and review. This is valuable because it reveals the real workflow before automating it.
- The October beta should use improved protocols, better documentation, some partial automation, and one or two controlled school/community cohorts.
- The January/February 2027 launch should be the first serious operational version, with reliable import, task assignment, review queues, quality snapshots, export manifests, and integration back to the Kindness Platform.
- Beyond January/February, the project should mature into a civic data production system with contributor skill ladders, advanced annotation pathways, steward roles, active review workflows, dataset release pipelines, and model validation.

## 1. The core question

The project should not be framed around the question, “Can volunteers label simple images?” That is too weak. A trivial task such as “is there a weed in this image?” may be useful as triage, but it is not enough to train a robot to act in the world. The stronger question is:

### Foundational question

Can non-experts, through structured apprenticeship and quality-controlled workflows, become reliable contributors to real-world usable datasets?

This distinction matters because the project is not merely an educational labelling exercise. It is attempting to create a civic data production capability: a system through which schools, adult volunteers, domain experts, data scientists, and reviewers can collectively produce datasets that are trustworthy enough for downstream machine-learning and robotics use.

## 2. What “real-world usable” means

A real-world usable dataset is not defined by volume alone. It is defined by whether the data can support a meaningful downstream model or operational system. For robotics, recycling, biodiversity, agriculture, and environmental monitoring, this usually requires more than image-level labels.

| Dataset goal              | Likely annotation needed                                | Why it matters                                                                                                      |
|---------------------------|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Image filtering / triage  | Image classification, bad image flag, uncertainty label | Filters out unusable or irrelevant captures and routes useful images into later queues.                             |
| Basic object presence     | Image classification                                    | Answers whether a target object or condition appears in the image. Useful but often not enough for physical action. |
| Object localisation       | Point annotation or bounding boxes                      | Tells a model where the target is in the image. This is often the first step toward robotic usefulness.             |
| Object detection          | Bounding boxes                                          | Enables models to locate weeds, crop plants, electrical items, batteries, circuit boards, or other targets.         |
| Precise shape or boundary | Polygons or segmentation masks                          | Needed when the exact object                                                                                        |

|                               |                                                                   |                                                                                                              |
|-------------------------------|-------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
|                               |                                                                   | outline, crop/weed boundary, material boundary, or action zone matters.                                      |
| Robotic action or disassembly | Points, boxes, keypoints, polygons, segmentation, sometimes video | Robots may need target centres, grasp points, no-go zones, component boundaries, or frame-to-frame tracking. |
| Video or moving systems       | Video annotation / tracking                                       | Needed for drone footage, tractor-mounted cameras, robotic footage, conveyor belts, or moving items.         |

The implication is that simple labelling tasks should be treated as the first rung of a production ladder, not as the final destination. They are useful for training, triage, routing, and identifying capable contributors. The long-term workflow must support progression toward point, box, polygon, segmentation, video, review, and steward-level tasks.

### 3. The apprenticeship model

The project should adopt an apprenticeship model. This means that volunteers do not all perform the same work forever. Instead, contributors progress through increasingly difficult task types as they develop competence, interest, and reliability.

| Level | Role / capability              | Typical tasks                                                                                          |
|-------|--------------------------------|--------------------------------------------------------------------------------------------------------|
| 0     | Orientation                    | Understand the project purpose, data ethics, and basic workflow.                                       |
| 1     | Capture contributor            | Collect images using KoboToolbox and the agreed capture protocol.                                      |
| 2     | Triage / classifier            | Classify images as usable, bad, unknown, relevant, not relevant, or broad category.                    |
| 3     | Point annotator                | Click weed centres, item centres, growth points, defects, or grasp points.                             |
| 4     | Box annotator                  | Draw bounding boxes around targets such as weeds, crops, batteries, circuit boards, phones, or motors. |
| 5     | Polygon annotator              | Trace object boundaries for high-value images where shape matters.                                     |
| 6     | Segmentation / video annotator | Produce masks or track objects across video frames.                                                    |
| 7     | Reviewer / steward             | Review others' work, catch errors, mentor contributors, and escalate difficult cases.                  |
| 8     | Expert-adjacent specialist     | A highly trained non-expert who can perform complex annotation reliably under protocol.                |

This ladder is important for two reasons. Operationally, it creates a way to route work to contributors with the right skill level.

Psychologically, it gives volunteers a path toward mastery, status, and meaningful contribution. A mature civic data commons should not

rely only on casual microtasks. It should build a community of increasingly capable contributors.

#### 4. The central scaling challenge: expert knowledge into volunteer training

A major bottleneck is likely to be the translation of expert judgement into documentation and training that non-experts can follow. A domain expert may know intuitively where a plant boundary is, when a weed is identifiable, whether a component contains valuable material, or when an image is too ambiguous. Volunteers need this converted into concrete, visual, repeatable rules.

For every serious annotation task, the project should produce a Task Specification Pack. This pack should be versioned, tested, and improved after each pilot.

| Component         | Purpose                                                                                                                                     |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Task purpose      | Explain what downstream model or robotic action this annotation supports.                                                                   |
| Object definition | Define exactly what counts as the object, component, weed, crop, material, defect, or target.                                               |
| Annotation type   | State whether the task is classification, point, box, polygon, mask, or video track.                                                        |
| Inclusion rules   | Define when the target should be labelled.                                                                                                  |
| Exclusion rules   | Define when the target should not be labelled or should be marked uncertain.                                                                |
| Boundary rules    | For boxes, polygons, and masks, define whether to include visible parts only, hidden parts, shadows, gaps, overlapping objects, stems, etc. |
| Ambiguity rules   | Tell volunteers when to mark uncertain, skip, escalate, or send to review.                                                                  |
| Good examples     | Show correct annotations.                                                                                                                   |
| Bad examples      | Show common mistakes.                                                                                                                       |
| Edge cases        | Show difficult cases such as occlusion, overlap, blur, shadows, fragments, and confusing backgrounds.                                       |
| Practice set      | Provide practice tasks with feedback before volunteers work on production data.                                                             |
| Review rubric     | Give reviewers criteria for accepting, rejecting, correcting, or escalating work.                                                           |

This documentation process may become one of Citizen AI's most important capabilities. The system is not just a software stack; it is a repeatable method for converting expert judgement into distributed dataset labour.

#### 5. Proposed technology architecture

The recommended architecture is a modular stack. The project should reuse strong open-source tools, but it should own the workflow brain itself.

##### Architectural rule

The Citizen AI Dataset Orchestrator should be the source of truth for dataset workflow state. Kobo, CVAT, Label Studio, FiftyOne, and the Kindness Platform can each perform specific functions, but none of them should be the canonical workflow system.

| Layer                  | Recommended technology                   | Role                                                                                         |
|------------------------|------------------------------------------|----------------------------------------------------------------------------------------------|
| Capture                | KoboToolbox                              | Field image and metadata collection. Keep using it while it works at scale.                  |
| Dataset Orchestrator   | Django + Django REST Framework           | The control plane for projects, images, tasks, states, review, quality metrics, and exports. |
| Database               | PostgreSQL                               | Canonical relational state, metadata, events, tasks, annotations, reviews, releases.         |
| Background jobs        | Celery + Redis                           | Imports, downloads, thumbnail generation, exports, quality snapshots, webhook retries.       |
| Image storage          | S3-compatible object storage             | Raw images, thumbnails, redacted images, release packages, manifests.                        |
| Simple labelling UI    | Native Django/HTMX or React UI           | High-volume classification, triage, review, and possibly point tasks.                        |
| Advanced annotation    | CVAT                                     | Points, boxes, polygons, segmentation, video, and complex computer-vision annotation.        |
| Exploratory annotation | Label Studio                             | Useful for experiments, but not recommended as long-term workflow backbone.                  |
| Visual QA              | FiftyOne                                 | Internal dataset inspection, model failure analysis, label review, metadata exploration.     |
| Release/versioning     | Manifest exports first; later DVC/lakeFS | Versioned dataset releases, dataset cards, QA summaries, train/validation/test splits.       |
| Platform integration   | Kindness API                             | Identity and role context in; completion and quality metrics out.                            |

## 6. Why build a Dataset Orchestrator?

The foundational technology decision should not be “Label Studio versus CVAT.” The foundational decision is that Citizen AI needs its own Dataset Orchestrator. This is because the core problem is not merely drawing labels. The core problem is assigning the right work to the right people, tracking provenance, managing review, measuring quality, and producing controlled dataset releases.

The Orchestrator should know:

- this image exists and has a canonical Citizen AI ID;
- it came from this Kobo submission and capture protocol version;
- it passed or failed capture QA;
- it is ready for classification, point annotation, box annotation, polygon annotation, review, or expert audit;
- this user was assigned this task;
- this annotation was submitted under this label guide version;
- this review agreed, disagreed, corrected, or escalated the annotation;
- this item entered or did not enter a release;
- this contributor has these quality metrics and skill levels;

- this completion or quality event should be sent back to Kindness.

Third-party tools should not own this state. If the project embeds the workflow inside CVAT or Label Studio, it may later struggle with custom queues, contributor skill ladders, Kindness integration, release provenance, quality scoring, and cross-domain dataset governance.

## 7. Core data model for the Orchestrator

The first version of the Orchestrator should not attempt to model everything perfectly. But it should establish the canonical entities that will remain stable across pilots.

| Entity                     | Purpose                                                                                                               |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------|
| DatasetProject             | Defines the dataset project, domain, status, protocol version, taxonomy version, and review policy.                   |
| CaptureSubmission          | Represents a Kobo submission or other source submission, including raw metadata and source IDs.                       |
| ImageItem                  | Canonical record for each image, with object storage URI, metadata, QA status, annotation status, and release status. |
| Contributor                | Local mirror of a Kindness user, including role context and project-specific skill profile.                           |
| AnnotationTask             | A unit of work assigned to a contributor, such as classification, point, box, polygon, or mask.                       |
| Annotation                 | The submitted label, point, box, polygon, mask, confidence, uncertainty reason, duration, and source tool.            |
| ReviewTask                 | A task asking another person to review a submitted annotation.                                                        |
| Review                     | Reviewer decision: agree, disagree, uncertain, bad image, corrected label, comment.                                   |
| DisagreementCase           | A queue item created when annotations conflict or quality thresholds are not met.                                     |
| Adjudication               | Steward or expert decision on a disputed or high-value item.                                                          |
| ContributorQualitySnapshot | Periodic quality metrics by user, project, task type, and role.                                                       |
| DatasetRelease             | A named versioned release with manifest, labels, metadata, QA summary, and dataset card.                              |
| DatasetEvent               | Append-only event log for auditability, metrics, debugging, and integration.                                          |

## 8. Workflow states

The Orchestrator should manage each image as it moves through a lifecycle. Not every image needs every stage, but the system should support these states.

- captured
- imported
- capture\_qa\_pending
- capture\_qa\_passed or capture\_qa\_failed
- annotation\_ready
- annotation\_in\_progress

- annotation\_complete
- review\_ready
- review\_in\_progress
- review\_complete
- disagreement or audit\_ready
- adjudicated
- release\_candidate
- released or excluded

Every state transition should create a DatasetEvent. This will make the system auditable, debuggable, and analyzable. It will also allow the project to calculate quality metrics and pass meaningful events back to the Kindness Platform.

## 9. CVAT's role in the stack

CVAT should be treated as a specialist annotation engine for spatial annotation, not as the whole dataset workflow system. It is useful when the task requires points, bounding boxes, polygons, segmentation, video tracking, or advanced computer-vision annotation. It is less suitable as the master system for civic task assignment, contributor progression, quality scoring, release state, or Kindness integration.

| Annotation type               | Should CVAT be used? | Comment                                                                                  |
|-------------------------------|----------------------|------------------------------------------------------------------------------------------|
| Image classification / triage | Usually no           | A native UI is likely simpler and less intimidating.                                     |
| Point annotation              | Possibly             | CVAT can do this, but a native point UI could also be built later.                       |
| Bounding boxes                | Yes                  | Good fit for CVAT once object localisation is required.                                  |
| Polygons                      | Yes                  | Good fit for advanced contributors and spatial tasks.                                    |
| Segmentation masks            | Yes                  | Use for trained stewards, experts, or high-value subsets.                                |
| Video tracking                | Yes                  | Use later for robot, drone, tractor, or conveyor footage.                                |
| Review / stewardship          | Mixed                | CVAT can support review, but Citizen AI should own the review state and quality metrics. |

The ideal integration pattern is:

- Citizen AI Orchestrator selects images and contributors based on project rules and skill level.
- Citizen AI creates or assigns a CVAT task/job for spatial annotation.
- Contributor completes the task in CVAT.
- CVAT exports or signals completion through API/webhook.
- Citizen AI imports the annotation result into its own data model.
- Citizen AI creates review, disagreement, quality, and release events as needed.

## 10. Kindness Platform API boundary

The wider Kindness Platform should remain the source of identity,

community, team, and high-level role context. The Dataset Orchestrator should consume that context but make its own assignment decisions. Kindness can say that a person is eligible to label or review. Citizen AI should decide which exact task that person sees.

| Direction              | Information passed                                                    | Purpose                                                                             |
|------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Kindness to Citizen AI | User ID, roles, teams, communities, eligibility flags                 | Allow Citizen AI to create a local contributor record and assign appropriate tasks. |
| Citizen AI to Kindness | Completion events, quality snapshots, accepted contribution summaries | Update progress, reputation, leagues, certificates, or contributor history.         |
| Citizen AI internal    | Image state, annotation state, review state, release state            | Must remain under Citizen AI control.                                               |

## 11. Pilot strategy

The project should not wait for the full Orchestrator before piloting. It should pilot the human workflow manually while building the technical stack in parallel. This is the best way to avoid building a system based on untested assumptions.

|                                                                                                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Pilot principle</b><br>Pilot the workflow manually before automating it. Build the Orchestrator around observed workflow, not imagined workflow. |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|

## 12. Summer pilot: adult Meetup / field pilot

The summer pilot should use adult volunteers, such as Meetup participants, and be explicitly framed as a workflow and process test. Volunteers should be told that some handoffs are manual because the project is deliberately testing capture, annotation, review, and training before automating the process.

| Pilot element   | Recommended design                                                                                      |
|-----------------|---------------------------------------------------------------------------------------------------------|
| Participants    | 20–30 adult volunteers.                                                                                 |
| Location        | Field site such as FFR or another domain-relevant site.                                                 |
| Capture tool    | KoboToolbox.                                                                                            |
| Annotation tool | CVAT or Label Studio depending on task; CVAT preferred for point/box/polygon testing.                   |
| Workflow glue   | Manual exports, spreadsheets, task links, manual assignment, manual result matching.                    |
| Main purpose    | Learn where the process breaks before automating it.                                                    |
| Expected output | Process report, technical workflow specification, revised training materials, automation priority list. |

## 13. Summer pilot workflow

1. Brief volunteers on the project purpose and explain that this is a process test.



2. Train volunteers on the capture protocol using visual examples and field checklist.
3. Capture images through KoboToolbox, with required metadata.
4. Export or import images and metadata manually into a working storage area.
5. Run capture QA: usable, bad image, privacy issue, duplicate, protocol failure, unclear target.
6. Create CVAT tasks manually or semi-manually for point annotation.
7. Train volunteers on point tasks and measure completion, time, confusion, and quality.
8. Create CVAT tasks for bounding boxes and repeat the same measurement process.
9. Offer polygon tasks as an advanced challenge and identify who can perform them well.
10. Run a small review exercise where volunteers review others' annotations.
11. Debrief volunteers and record what was confusing, enjoyable, tedious, or motivating.
12. Export annotations and manually calculate initial workflow and quality metrics.
13. Write a process report and a technical workflow specification.

#### 14. What the summer pilot should measure

| Area                  | Metrics / observations                                                                                                             |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Capture               | Usable image rate, metadata completeness, blur rate, duplicate rate, personal-data incidents, protocol compliance, time per image. |
| Training              | Which instructions were misunderstood, what examples were missing, how long volunteers needed, confidence vs accuracy.             |
| CVAT usability        | Login problems, time to first annotation, save/submit errors, zoom/pan confusion, wrong label selection, task completion.          |
| Point annotation      | Distance from reference point, wrong-object click rate, time per item, improvement after practice.                                 |
| Bounding boxes        | Overlap with reference boxes, missed objects, too-tight/too-loose boxes, time per image, repeated mistakes.                        |
| Polygons              | Completion rate, time per polygon, boundary quality, fatigue, whether a subset of volunteers becomes capable.                      |
| Review                | Reviewer agreement, false acceptance, false rejection, ability to spot ambiguity.                                                  |
| Volunteer progression | How many volunteers move from capture/classification to point, box, polygon, and review.                                           |
| Operational burden    | Manual steps, time required, errors introduced, automation priority.                                                               |

#### 15. Manual workflow debt log

Every manual step in the summer pilot should be recorded as “manual workflow debt.” This prevents the team from confusing a successful manually supported pilot with a scalable operational system.

| Manual step             | Question to answer                             | Possible decision                                  |
|-------------------------|------------------------------------------------|----------------------------------------------------|
| Export images from Kobo | How long did it take and what errors occurred? | Automate by October or January.                    |
| Rename/create image IDs | Did matching break between tools?              | Automate early; canonical image IDs are important. |
| Upload images to CVAT   | Was this slow or error-prone?                  | Semi-automate before October if repeated.          |
| Create CVAT tasks       | How many manual settings were needed?          | Automate or template by January.                   |
| Assign users to tasks   | Could this be done reliably with               | Manual for summer; orchestrator                    |

|                             |                                            |                                          |
|-----------------------------|--------------------------------------------|------------------------------------------|
|                             | spreadsheets?                              | by January.                              |
| Export CVAT results         | Were results easy to match back to images? | Automate import into Orchestrator.       |
| Calculate quality metrics   | How much data wrangling was needed?        | Manual for summer; automate by January.  |
| Route reviews/disagreements | Did review queues become confusing?        | Build queue states and review dashboard. |

## 16. October beta

The October beta should be a controlled run with one or two schools or comparable cohorts. The goal is not a full commercial-scale launch. The goal is to use the summer pilot learning to make the workflow much more reliable for a more structured group.

| By October, aim to have           | Notes                                                                           |
|-----------------------------------|---------------------------------------------------------------------------------|
| Improved capture protocol         | Based on summer capture failures and expert feedback.                           |
| Improved task specification packs | Especially for points, boxes, and introductory polygon tasks.                   |
| Kobo import partially automated   | At least enough to avoid fragile manual matching.                               |
| Canonical ImageItem records       | Every image should have a stable Citizen AI ID.                                 |
| Basic storage architecture        | Images copied out of Kobo into controlled storage.                              |
| Basic admin/steward screen        | View imported images, QA status, annotation readiness.                          |
| CVAT task workflow tested         | Manual or semi-automated is acceptable if reliable.                             |
| Basic quality metrics             | Capture pass rate, task completion, simple annotation metrics, review outcomes. |
| Support runbook                   | What to do if login, upload, annotation, or export fails.                       |
| Fallback plan                     | Pre-collected image sets and manual recording if a tool fails.                  |

## 17. January/February 2027 operational launch

January or February should be treated as the first serious operational launch. By this stage, the Orchestrator should no longer be a theoretical architecture. It should perform the key control-plane functions needed to run a real dataset workflow at larger scale.

| Capability           | Expected state by Jan/Feb                                                                         |
|----------------------|---------------------------------------------------------------------------------------------------|
| Kobo ingestion       | Automated or reliable semi-automated import from Kobo using stable source IDs and metadata.       |
| Image custody        | Images copied into object storage with canonical Citizen AI IDs and thumbnails.                   |
| Task queues          | Images can be routed to capture QA, classification, point, box, polygon, review, or audit queues. |
| Contributor profiles | Basic skill levels by task type: capture, classification, point, box, polygon, review.            |
| CVAT integration     | At minimum reliable task creation/export/import workflow; ideally API-assisted.                   |
| Review workflow      | Review tasks, disagreement cases, and steward queues are tracked.                                 |
| Quality snapshots    | Basic user/team/cohort metrics calculated and available for feedback/API.                         |
| Event log            | All key actions logged for provenance and debugging.                                              |

|                           |                                                                          |
|---------------------------|--------------------------------------------------------------------------|
| Exports                   | CSV/JSON/image manifest exports for data science and release candidates. |
| Kindness integration      | Identity/role handoff in; completion and quality events out.             |
| Operational documentation | Deployment, backup, support, pilot runbook, and data handling notes.     |

## 18. Beyond January/February

After the first operational launch, the system should mature toward a world-class civic data production stack. Later capabilities should be added when the workflow has justified them.

| Later capability                | Why it matters                                                                                                            |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Native high-volume labelling UI | Simplifies classification, triage, and review for large volunteer cohorts.                                                |
| Better CVAT automation          | Create tasks, assign jobs, import annotations, and manage spatial workflows programmatically.                             |
| Contributor progression engine  | Unlock advanced tasks based on training, quality, and consistency.                                                        |
| Advanced quality scoring        | Use point distance, box IoU, polygon IoU/Dice, review alignment, expert override rates, and calibration items.            |
| Model-assisted pre-labelling    | Use baseline models to pre-label or route items, with humans verifying and correcting.                                    |
| Active learning workflows       | Prioritise items where model uncertainty or data value is highest.                                                        |
| FiftyOne QA integration         | Inspect datasets, labels, predictions, failure modes, outliers, and metadata bias.                                        |
| Dataset release pipeline        | Versioned releases with manifests, dataset cards, QA summaries, train/validation/test splits, and baseline model results. |
| DVC/lakeFS or equivalent        | Reproducible dataset versioning once release discipline matures.                                                          |
| Public dataset portal           | Make selected public-good releases discoverable and well documented.                                                      |

## 19. Team structure

The project should now split into two linked teams: one focused on human workflow discovery and one focused on technical orchestration. This prevents two opposite failures: building software nobody needs, or learning a lot from pilots without industrialising the process.

| Team                               | Lead profile                                  | Responsibilities                                                                                                                 |
|------------------------------------|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Pilot & Workflow Team              | Data scientist / project manager              | Run pilots, observe workflow, document failures, refine protocols, coordinate volunteers, gather metrics, produce pilot reports. |
| Orchestrator & Infrastructure Team | Strong technical volunteer / engineering lead | Build Django/Postgres Orchestrator, import/storage/event models, CVAT experiments, APIs, exports, deployment, and automation.    |
| Domain Expert Interface            | Domain expert plus workflow translator        | Convert expert judgement into task specification packs, edge                                                                     |

|                        |                                   |                                                                                                     |
|------------------------|-----------------------------------|-----------------------------------------------------------------------------------------------------|
|                        |                                   | cases, practice sets, and review rubrics.                                                           |
| Data Science / QA Team | Data scientists and statisticians | Analyse pilot outputs, compute quality metrics, inspect label reliability, guide release standards. |

## 20. Weekly operating rhythm

The two-track structure only works if the teams communicate frequently. A weekly or fortnightly review should ask:

- What did the pilot/workflow team learn from real users?
- Which manual steps were painful and should be automated?
- Which assumptions changed?
- Which task definitions or training materials need revision?
- Which data model or API changes are required?
- What can remain manual until October?
- What must be automated before January/February?
- Which risks are increasing?
- Which workflows are now stable enough to encode in software?

## 21. Technical roadmap

### 21.1 Now to summer pilot

- Set up Django/Postgres Orchestrator skeleton.
- Create initial models: Project, Contributor, CaptureSubmission, ImageItem, Task, Annotation, Review, DatasetEvent.
- Set up object storage and image path conventions.
- Prototype Kobo import and image copying.
- Create or refine CVAT deployment/test instance.
- Test point, box, and polygon workflows in CVAT with a few non-team users before the full pilot.
- Create manual workflow scripts where useful, but do not over-automate yet.
- Create pilot observation templates and manual workflow debt log.
- Prepare task specification packs for capture, point, box, and polygon tasks.

### 21.2 Summer to October

- Incorporate pilot findings into task documentation and training materials.
- Automate or semi-automate the most painful import/export/matching steps.
- Stabilise canonical image IDs and object storage.
- Build a basic capture QA/admin screen.
- Build basic task state tracking.
- Implement basic quality metrics and export reports.
- Create a reliable support/runbook for school or cohort beta.
- Decide which tasks use native UI and which use CVAT in the October beta.
- Add minimal Kindness session handoff if needed for the beta.

### 21.3 October to January/February

- Build reliable task queue management.
- Improve CVAT integration through API or robust import/export scripts.
- Add contributor skill profiles and simple task eligibility.
- Add review and disagreement queues.

- Add quality snapshots by contributor, team/cohort, and task type.
- Add event pushback to Kindness for completion and quality metrics.
- Add release candidate export: images, labels, metadata, manifest, QA summary, dataset card draft.
- Integrate FiftyOne into internal QA workflow.
- Harden deployment, backup, monitoring, and documentation.

## 21.4 After January/February

- Develop native labelling/review UI for high-volume classification and triage.
- Deepen CVAT automation for spatial workflows.
- Build robust contributor progression and promotion rules.
- Implement more sophisticated reliability modelling and audit sampling.
- Add model-assisted annotation and active learning.
- Formalise dataset release/versioning pipeline.
- Build public dataset documentation and release portal when appropriate.
- Extend architecture to additional domains beyond the first pilot.

## 22. Risk register

| Risk                                | Why it matters                                                             | Mitigation                                                                                                                  |
|-------------------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Over-simplifying tasks              | The dataset may not be useful for real robotics or ML.                     | Treat simple tasks as triage/training, not the endpoint. Progress to points, boxes, polygons where needed.                  |
| Overbuilding before learning        | The team may build the wrong system.                                       | Run summer pilot manually and use it to define automation priorities.                                                       |
| Manual pilot hides scaling cost     | A pilot can appear successful only because skilled people did hidden work. | Maintain a manual workflow debt log and classify each step as keep manual, automate by October, or automate by January.     |
| CVAT complexity                     | Volunteers may struggle with professional annotation UI.                   | Test with points/boxes first, provide task-specific guides, use CVAT for advanced tracks, and keep Orchestrator in control. |
| Poor expert-to-training translation | Volunteers cannot produce consistent labels without clear rules.           | Create task specification packs with good/bad examples, edge cases, and review rubrics.                                     |
| Image capture quality too low       | Bad input data limits downstream usefulness.                               | Use field checklists, protocol testing, capture QA, and rapid feedback.                                                     |
| Annotation backlog                  | Many images may be captured faster than polygons can be produced.          | Use queues and contributor progression; not every image needs every annotation type.                                        |
| Volunteer burnout                   | Advanced annotation can feel like unpaid drudgery.                         | Create progression, status, feedback, monthly sprints, and visible impact.                                                  |
| Tool lock-in                        | Workflow state trapped in CVAT/Label Studio/Kobo.                          | Make Citizen AI Orchestrator the source of truth.                                                                           |
| Delayed infrastructure              | Manual pilots succeed but automation is not ready for January.             | Run pilot and technical build in parallel, with weekly feedback loops.                                                      |

## 23. What success looks like

The project should not define success as “everything worked perfectly.” Early pilots are successful if they reveal what is feasible,

what is hard, what needs automation, and what training is required. By January/February, success should look like a reliable operational workflow that can produce evidence-backed dataset releases.

| Stage                   | Success looks like                                                                                                                                     |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Summer pilot            | Real volunteers complete capture and annotation tasks; the team measures feasibility, failure modes, progression, and automation needs.                |
| October beta            | A controlled cohort completes a more reliable workflow with improved documentation, partial automation, support runbook, and basic quality metrics.    |
| January/February launch | The Orchestrator supports import, image custody, task routing, review, quality snapshots, exports, and Kindness integration for a serious dataset run. |
| Longer term             | Citizen AI produces versioned, documented, quality-controlled public-good datasets with measurable reliability and downstream model validation.        |

## 24. Final recommendation

The recommended path is to proceed with a dual-track approach:

- Run the summer pilot manually or semi-manually to discover the real workflow.
- Build the Django/Postgres Dataset Orchestrator in parallel, focusing first on stable entities, storage, events, imports, tasks, and exports.
- Use CVAT for spatial annotation tasks, but keep Citizen AI as the source of truth.
- Use Kobo for capture while it continues to work, but copy images and metadata into Citizen AI-controlled storage.
- Use the summer pilot to identify the human, documentation, and workflow bottlenecks.
- Use the October beta to test a more stable and partially automated system.
- Use January/February 2027 as the first serious operational launch target.
- Continue maturing the system beyond January with progression, advanced QA, release versioning, model validation, and expanded domains.

### Guiding principle

Own the workflow. Reuse the tools. Train non-experts through a real progression path. Automate only what the pilots prove is worth automating. Build toward datasets that are useful, auditable, versioned, and trusted.

## Appendix A: Example event types

| Event type                | Meaning                                                         |
|---------------------------|-----------------------------------------------------------------|
| kobo_submission_imported  | A capture submission was imported from Kobo.                    |
| image_item_created        | A canonical ImageItem was created.                              |
| image_failed_capture_qa   | Image was rejected or flagged during capture QA.                |
| annotation_task_assigned  | A contributor was assigned an annotation task.                  |
| annotation_submitted      | Contributor submitted a label, point, box, polygon, or mask.    |
| review_task_assigned      | A review task was created.                                      |
| review_submitted          | Reviewer submitted a decision.                                  |
| disagreement_created      | Conflicting annotations or reviews created a disagreement case. |
| adjudication_completed    | A steward or expert resolved a case.                            |
| quality_snapshot_updated  | Contributor or cohort quality metrics were recalculated.        |
| release_candidate_created | A dataset release candidate was generated.                      |
| item_added_to_release     | An item was included in a release.                              |

|                            |                                                                           |
|----------------------------|---------------------------------------------------------------------------|
| item_excluded_from_release | An item was excluded from a release.                                      |
| kindness_event_sent        | Completion or quality information was sent back to the Kindness Platform. |

## Appendix B: Example quality metrics by annotation type

| Task type               | Quality measures                                                                                                   |
|-------------------------|--------------------------------------------------------------------------------------------------------------------|
| Capture                 | Protocol compliance, metadata completeness, blur rate, duplicate rate, usable image rate, privacy issue rate.      |
| Classification          | Agreement with reference labels, confusion matrix, uncertainty usage, overconfidence rate, time per item.          |
| Point annotation        | Mean distance from reference point, percentage within acceptable radius, wrong-object click rate.                  |
| Bounding boxes          | Intersection over Union, missed objects, false objects, too-tight or too-loose patterns.                           |
| Polygons / segmentation | IoU, Dice coefficient, boundary error, inclusion of wrong pixels, exclusion of target pixels, time per annotation. |
| Review                  | Alignment with expert/steward decisions, false accept rate, false reject rate, ability to identify ambiguity.      |

## Appendix C: Example architecture diagram in text

KoboToolbox

- > Kobo import worker
- > Citizen AI Dataset Orchestrator (Django + DRF + Postgres)
- > Object storage for images and releases
- > Native UI for simple tasks and/or CVAT for spatial tasks
- > Review, disagreement, quality, and release workflows
- > FiftyOne for internal visual QA
- > Kindness API for identity in and completion/quality events out

## Appendix D: Immediate next actions

14. Confirm the summer pilot domain, site, and approximate date.
15. Assign the Pilot Operations & Workflow Lead.
16. Assign the Orchestrator & Infrastructure technical lead.
17. Create the summer pilot evaluation framework before recruiting participants.
18. Prepare task specification packs for capture, point, box, and polygon tasks.
19. Run a small CVAT usability test with 3–5 non-team users before the full pilot.
20. Begin the Django/Postgres Orchestrator skeleton immediately.
21. Create the manual workflow debt log and use it during the summer pilot.
22. Schedule weekly feedback meetings between the pilot and technical teams.
23. After the summer pilot, classify each manual step as: keep manual for now, semi-automate by October, must automate by January, or remove.